

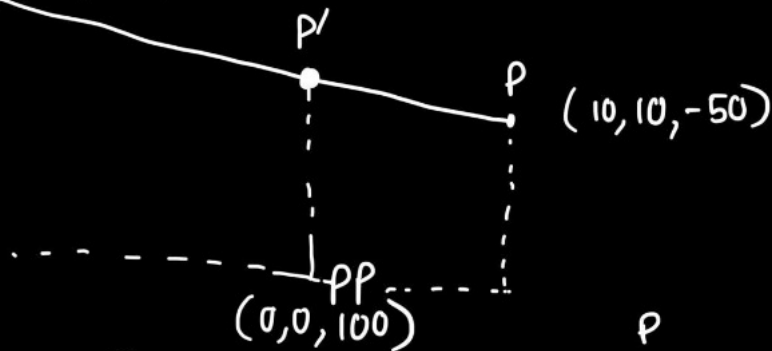
CoP (30, 20, 120)

$$q_x = 30 - 0 = 30$$

$$q_y = 20 - 0 = 20$$

$$q_z = 120 - 100 = 20$$

$$z_p = 100$$



$$\begin{bmatrix} 1 & 0 & -\frac{30}{20} & (100 \cdot \frac{30}{20}) \\ 0 & 1 & -\frac{20}{20} & (100 \cdot \frac{20}{20}) \\ 0 & 0 & -100/20 & (100 + \frac{10000}{20}) \\ 0 & 0 & -1/20 & (1 + \frac{100}{20}) \end{bmatrix} \begin{bmatrix} 10 \\ 10 \\ -50 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} 235 \\ 160 \\ 850 \\ 8.5 \end{bmatrix} = \begin{bmatrix} 10 + 75 + 150 \\ 10 + 50 + 100 \\ 250 + 600 \\ 2.5 + 6 \end{bmatrix} = \begin{bmatrix} 1 & 0 & -1.5 & 150 \\ 0 & 1 & -1 & 100 \\ 0 & 0 & -5 & 600 \\ 0 & 0 & -0.05 & 6 \end{bmatrix} \begin{bmatrix} 10 \\ 10 \\ -50 \\ 1 \end{bmatrix}$$

→ Normalise → $(\div 8.5)$

$$\begin{bmatrix} 27.6 \\ 18.8 \\ 100 \\ 1 \end{bmatrix}$$

$$Q1) C_x = (0.2, 0.35, 0.4)$$

$$+ (0.7 - 0.2, 0.7 - 0.35, 0.7 - 0.4) \left[\frac{80 - 35}{90 - 35} \right]$$

$$= (0.2, 0.35, 0.4) + \left[(0.5, 0.35, 0.3) \times \frac{9}{11} \right]$$

$$= (0.609, 0.636, 0.645)$$

$$C_m = (0.25, 0.55, 0.45) + (0.609 - 0.25, 0.636 - 0.55, 0.645 - 0.45)$$

$$\left[\frac{170 - 120}{220 - 120} \right] / \frac{50}{100} / \frac{1}{2}$$

$$= (0.4295, 0.593, 0.5475)$$

$$\text{In RGB} = (0.5705, 0.407, 0.4525)$$

$$\text{HSV } C_{\max} = 0.5705$$

$$C_{\min} = 0.407$$

$$\text{diff} = 0.1635$$

$$H = \left(\frac{0.407 - 0.4525}{0.1635} \times 60 \right) = -16.697 + 360$$

$$S = \frac{0.1635}{0.5705} = 0.289$$

$$= 343.3^\circ$$

$$V = C_{\max} = 0.5705$$

1. HLS model

$$C = V \times S = 0.71 \times 0.45 = 0.3195$$

$$m = 0.71 - 0.3195 = 0.3905$$

$$X = 0.3195 \times (1 - |(1.067 - 1)|) = 0.298$$

$$180 \leq H < 240 \rightarrow R=0, G=X, B=C$$

$$R=0, G=0.298, B=0.3195$$

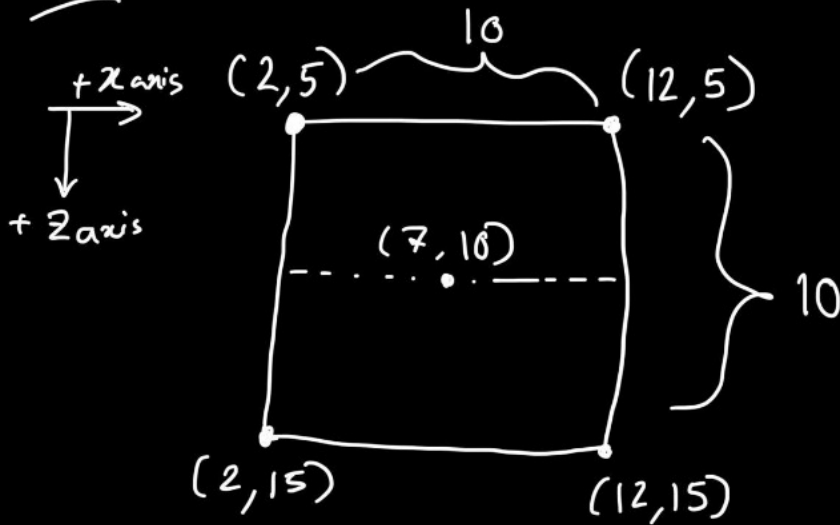
$$+m \rightarrow R=0.3905, G=0.6885, B=0.71$$

$$\begin{aligned} CMY &= (1 - 0.3905, 1 - 0.6885, 1 - 0.71) \\ &= (0.6095, 0.3115, 0.29) \end{aligned}$$

2. Parallel projection definition with diagram

$$\begin{matrix} P' \\ \begin{bmatrix} 50 \\ 15 \\ 72 \\ 1 \end{bmatrix} \end{matrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 15 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \times \begin{matrix} P \\ \begin{bmatrix} 50 \\ 32 \\ 72 \\ 1 \end{bmatrix} \end{matrix}$$

Q2 Mirror $\rightarrow$ Diffuse Reflektion = 0



$$\vec{P} = (7, 0, 10) \quad \vec{L} = (1, 1, 4) \quad \hat{n} = \hat{j}$$

$$\begin{aligned} \vec{L} &= (1, 1, 4) - (7, 0, 10) \\ &= (-6, 1, -6) \end{aligned}$$

$$\hat{L} = \frac{-6i + j - 6k}{\sqrt{73}}$$

$$\hat{n} \cdot \hat{L} = \frac{1}{\sqrt{73}}$$

$$\begin{aligned} R &= \frac{2}{\sqrt{73}} \hat{j} - \left(\frac{-6i + j - 6k}{\sqrt{73}} \right) \\ &= \frac{6i + j + 6k}{\sqrt{73}} \end{aligned}$$

$$\begin{aligned} V &= (3, 2, 6) - (7, 0, 10) \\ &= (-4, 2, -4) \end{aligned}$$

$$\hat{V} = \frac{-4i + 2j - 4k}{6}$$

$$\begin{aligned} R \cdot V &= \frac{-24}{6\sqrt{73}} + \frac{2}{6\sqrt{73}} - \frac{24}{6\sqrt{73}} \\ &= \frac{-23}{3\sqrt{73}} \end{aligned}$$

$$\begin{aligned} \text{Specular} &= 10 \times 0.5 \times \left[\left(\frac{-23}{3\sqrt{73}} \right)^5, 0 \right]_{\max} \\ &= \underline{0} \end{aligned}$$

$$C_y = (0.25, 0.55, 0.45) + (0.7, 0.7, 0.7 - 0.25, 0.55, 0.45) \left[\frac{85-80}{90-80} \right] \\ = (0.475, 0.625, 0.575) \quad (0.5)$$

$$C_m = (0, 0, 0) + (0.475, 0.625, 0.575) \left[\frac{80-40}{85-40} \right] \\ = 0.422, 0.556, 0.511$$

$$\text{In RGB} = (0.578, 0.444, 0.489)$$

HSV

$$c_{max} = 0.578$$

$$c_{min} = 0.444$$

$$\text{diff} = 0.134$$

$$H = \frac{0.444 - 0.489}{0.134} \times 60 = -20.15 + 360 = \underline{339.85^\circ}$$

$$S = \frac{0.134}{0.578} = \underline{0.232}$$

$$V = c_{max} = \underline{0.578}$$

OR